Certify-Join: Full Proofs

Technical report accompanying *Certify-Join: Budgeted Certification and Incremental Maintenance of Aggregation-Safe Join Graphs*

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Abstract

This report contains the definitions and complete proofs of the formal results stated in the paper. Theorem numbers below differ from those in the paper; Table 1 gives the correspondence. Where a statement in the paper is a simplification, the precise hypotheses are stated here and the differences are pointed out in remarks.

Paper label	Content	This report
`thm:hard`	NP-hardness and inapproximability of offline BJC	Theorem 8, Corollaries 10–12
`thm:tree`	Tree-structured case: pseudo-polynomial DP and FPTAS	Theorem 13, Proposition 15
(cited only)	Optimal probe order within a bundle	Proposition 16
`thm:sound`	Soundness of exact and sampled certificates	Theorems 3, 18
`thm:lower`	Linear lower bound for certifying a key atom	Theorem 21
`thm:maint`	Exact incremental maintenance; necessity of state	Theorem 24, Propositions 27, 28
`thm:dir`	Certificates are directional	Theorem 5, Proposition 7
`lem:patheq`	Answer invariance iff path equivalence	Theorem 30, Corollary 31

Table 1: Correspondence between the paper and this report.

1 Preliminaries

Instances. A database D consists of finite tables. A table T is a bag of rows over a set $\mathrm{attr}(T)$ of attributes; every row carries an identity, so two rows may agree on all attributes and still be distinct. Values are drawn from a domain extended with NULL. For a list $X = (A_1, \ldots, A_\ell)$ of attributes and a row t we write $t[X] = (t[A_1], \ldots, t[A_\ell])$, and we say that $t[X]$ is *total* if no component is NULL. Two lists are compared positionally. Following SQL, an equality predicate $t[X] = s[Y]$ is true only if both sides are total and equal.

Atoms. Let S be a table and Y a list of its attributes.

- The *key atom* $\mathsf{U}(S, Y)$ holds on D if there are no two distinct rows $s \neq s'$ of S with $s[Y]$ total and $s[Y] = s'[Y]$. (Rows whose Y is not total are exempt, as for SQL `UNIQUE`; they never match an equality predicate.)

- The *temporal key atom* $\mathsf{U}(S, Y; fr, to)$, for validity attributes fr, to of S that are disjoint from Y, holds if there are no two distinct rows $s \neq s'$ with $s[Y]$ total, $s[Y] = s'[Y]$, and overlapping closed validity intervals, i.e. $[s.fr, s.to] \cap [s'.fr, s'.to] \neq \emptyset$. (A row with a NULL bound has an empty interval.)

- The *inclusion atom* $\mathsf{I}(R.X \subseteq S.Y)$, for lists X of R and Y of S of equal length, holds if every row $r \in R$ with $r[X]$ total has a *partner* $s \in S$ with $s[Y] = r[X]$. A row of R with total X and no partner is an *orphan*; the *orphan rate* is the number of orphans divided by $|R|$.

- The *temporal inclusion atom* $\mathsf{I}(R.X \subseteq S.Y \mathbin{@} d)$, for a date attribute d of R, holds if every $r \in R$ with $r[X]$ total and $r.d \neq \mathsf{NULL}$ has a partner s with $s[Y] = r[X]$ and $s.fr \leq r.d \leq s.to$.

Edges and the join predicate. A candidate edge $e = (R.X \to S.Y)$, optionally temporal with date d of R and validity attributes of S, has the join predicate $\theta_e(r, s) \equiv \big(r[X] = s[Y]\big)$, extended by $s.fr \leq r.d \leq s.to$ in the temporal case. Its atoms are $\mathsf{U}(S, Y)$ (temporal: $\mathsf{U}(S, Y; fr, to)$) and $\mathsf{I}(R.X \subseteq S.Y)$ (temporal: $\mathsf{I}(R.X \subseteq S.Y \mathbin{@} d)$). The edge is *safe* on D if both atoms hold. All edges that enter the same (S, Y) share the key atom.

Inference rules. $\mathcal{R}$ consists of the following rules and their contrapositives (which derive falsity):

- *superkey*: $\mathsf{U}(S, Y) \Rightarrow \mathsf{U}(S, Y')$ whenever $Y \subseteq Y'$ as sets of attributes, and likewise for temporal key atoms with the same validity attributes;

- *transitivity*: $\mathsf{I}(R.X \subseteq S.Y) \wedge \mathsf{I}(S.Y \subseteq T.Z) \Rightarrow \mathsf{I}(R.X \subseteq T.Z)$ for non-temporal atoms.

Lemma 1 (Soundness of $\mathcal{R}$; cf. the inclusion-dependency axioms of [4])**.** *If the premises of a rule of $\mathcal{R}$ hold on D, its conclusion holds on D. Consequently, every atom derived by $\mathcal{R}$ from atoms that hold on D holds on D, and every atom refuted by a contrapositive from true and false atoms of D is false on D.*

Proof. Superkey: if $s \neq s'$ had $s[Y']$ total and $s[Y'] = s'[Y']$, then $s[Y]$ is total and $s[Y] = s'[Y]$ because $Y \subseteq Y'$, contradicting $\mathsf{U}(S, Y)$; the temporal case adds the same interval condition on both sides. Transitivity: let $r \in R$ with $r[X]$ total. By the first premise there is $s \in S$ with $s[Y] = r[X]$, so $s[Y]$ is total, and by the second premise there is $t \in T$ with $t[Z] = s[Y] = r[X]$. A contrapositive $\neg C \wedge P_1 \Rightarrow \neg P_2$ of a sound rule $P_1 \wedge P_2 \Rightarrow C$ is sound by propositional logic. The consequences follow by induction on the length of a derivation. □

Join trees. A *join tree* J rooted at a table F is a rooted tree whose nodes are pairwise distinct tables, with root F. Each non-root node C with parent P is reached by a *hop* $h = (P.X_h \to C.Z_h)$ whose predicate is $\theta_h(p, c) \equiv p[X_h] = c[Z_h]$ (for a temporal hop, additionally $c.fr \leq p.d \leq c.to$). A hop traverses a candidate edge e either *forward* ($P = R_e$, $C = S_e$, $X_h = X_e$, $Z_h = Y_e$) or *in reverse* ($P = S_e$, $C = R_e$, $X_h = Y_e$, $Z_h = X_e$; reverse hops are non-temporal). Fix an order $h_1, \ldots, h_k$ of the hops in which every parent precedes its children. The *left-join result* $F \ltimes J$ is defined by $Q_0 = F$ and $Q_i = Q_{i-1} \ltimes_{\theta_{h_i}} C_i$, where for each tuple $q \in Q_{i-1}$ with matching set $M_i(q) = \{c \in C_i : \theta_{h_i}(q[P_i], c)\}$ the result contains $q \circ c$ for every $c \in M_i(q)$ if $M_i(q) \neq \emptyset$ and $q \circ \mathsf{NULL}$ otherwise. Here $q[P_i]$ is the P_i-component of q, and a NULL-padded component matches nothing. The result does not depend on the chosen order. We say that hop h is *covered* by a set Σ of atoms if Σ contains $\mathsf{U}(C, W)$ for some $W \subseteq Z_h$, or, if h is temporal, $\mathsf{U}(C, W; fr, to)$ for some $W \subseteq Z_h$. A forward hop over e is covered by the atoms of e.

Workload and BJC. A query $q = (F_q, D_q, w_q)$ has an anchor table F_q, a set D_q of required tables and a weight $w_q \geq 0$. Each atom has a cost $c(a) \geq 0$ and a prior $p(a) \in [0, 1]$. Given a set A of atoms known to be true (probed, or derived by $\mathcal{R}$), an edge is *certified* if both its atoms are in A, and q is *certified* if every table of D_q is reachable from F_q in the directed graph of certified edges (edges oriented $R \to S$). In the *offline* problem the truth of every atom is known, and we seek a set P of true atoms with $c(P) \leq B$ maximizing the weight of queries certified by $A = \mathrm{Cl}_{\mathcal{R}}(P)$, the closure of P under the positive rules. (Probing a false atom only costs budget and certifies nothing, so an optimal offline policy probes true atoms only.) The decision version asks whether weight at least K is achievable.

2 Soundness of exact certificates (paper: Theorem `thm:sound`(a))

Lemma 2 (Multiplicity)**.** *Every tuple $q \in Q_{i-1}$ produces exactly* $\max(1, |M_i(q)|)$ *tuples of Q_i. In particular $|F \rtimes J| \geq |F|$. Equality holds if and only if every tuple of every Q_{i-1} has at most one match at hop h_i.*

Proof. Immediate from the definition of the left outer join. □

Theorem 3 (Exact certificates are sound)**.** *Let J be a join tree rooted at F and suppose that every hop of J is covered by a set of atoms that hold on D (for example, all hops are forward and all atoms of their edges were probed exactly or derived by $\mathcal{R}$ from probed atoms). Then:*

(i) There are functions $\varphi_T : F \to T \cup \{\bot\}$, one per table T of J, with $\varphi_F(r) = r$, such that $F \rtimes J$ is the bag $\{\{(\varphi_T(r))_{T \in J} : r \in F\}\}$, where $\bot$ stands for a NULL-padded component. In particular $|F \rtimes J| = |F|$ and every row of F occurs exactly once.

(ii) For every selection predicate σ on F, every list G of grouping attributes of tables of J, and every aggregate $\mathrm{agg} \in \{$`COUNT(*)`, `COUNT`, `SUM`, `AVG`, `MIN`, `MAX`$\}$ *over a measure μ of F, the value of group g in* `SELECT G, agg(μ) FROM F ⋊ J WHERE σ GROUP BY G` *equals* $\mathrm{agg}\{\{r[\mu] : r \in F,\ \sigma(r),\ \gamma(r) = g\}\}$*, where $\gamma(r)$ is the value of G on $(\varphi_T(r))_T$. Every row of F thus contributes exactly once, to exactly one group.*

(iii) If moreover every hop is forward and the inclusion atom of every edge of J holds, then for every $r \in F$ and every non-root C with parent P: $\varphi_C(r) = \bot$ if and only if $\varphi_P(r) = \bot$, or $\varphi_P(r)[X_h]$ is not total, or (temporal hop) $\varphi_P(r).d =$ NULL. That is, a fact row falls into a NULL group only through a NULL reference, never through a dangling one.

Proof. By Lemma 1, all atoms in the certificate hold on D. (i) Induction on i. For $i = 0$, $Q_0 = F$ and φ_F is the identity. Assume $Q_{i-1} = \{\{(\varphi_T(r))_{T \in J_{i-1}} : r \in F\}\}$ for the subtree J_{i-1} formed by the first $i - 1$ hops. Let $h_i = (P.X \to C.Z)$ be covered by a key atom on C with list $W \subseteq Z$. For q derived from r, if $p = \varphi_P(r) = \bot$ then $M_i(q) = \emptyset$. Otherwise any $c, c' \in M_i(q)$ satisfy $c[Z] = p[X] = c'[Z]$ with this value total, hence $c[W] = c'[W]$ total. A non-temporal $\mathsf{U}(C, W)$ then forces $c = c'$. If the hop is temporal and covered by $\mathsf{U}(C, W; fr, to)$, the intervals of c and c' both contain $p.d$, so they overlap, and the temporal key atom forces $c = c'$. Thus $|M_i(q)| \leq 1$; define $\varphi_C(r)$ as its unique element or $\bot$. By Lemma 2 q produces exactly one tuple, which proves the claim for i. (ii) By (i) the rows of the `FROM` clause are in bijection with the rows of F, the `WHERE` clause keeps those with $\sigma(r)$, and the row derived from r is placed in group $\gamma(r)$. Each aggregate is a function of the bag of measure values in a group, and that bag is $\{\{r[\mu] : \sigma(r), \gamma(r) = g\}\}$. (iii) If $\varphi_P(r) = p \neq \bot$, $p[X]$ is total and (temporal) $p.d \neq \mathsf{NULL}$, then p is a row of $P = R_e$ with total

X_e, and $\mathsf{I}(R_e.X_e \subseteq S_e.Y_e)$ provides a partner $c \in C = S_e$ satisfying $\theta_h(p, c)$; hence $\varphi_C(r) = c \neq \bot$. The converse direction is immediate from the definition of θ_h. □

Remark 4. `MIN` and `MAX` are insensitive to duplication but not to mis-assignment, which is why (iii) matters for them as well. Part (iii) is stated for forward hops only, because an inclusion atom constrains the referencing side of an edge.

3 Certificates are directional (paper: Theorem `thm:dir`)

Theorem 5 (Directionality)**.** *Let J be a join tree rooted at F and let Σ be any set of key and inclusion atoms over the tables of J (for instance, the atoms of the candidate edges used by J, plus any additional key atoms). The following are equivalent:*

(a) $|F \rtimes J| = |F|$ on every instance D that satisfies Σ;

(b) every hop of J is covered by Σ.

Moreover, if some hop is not covered then $\sup\{|F \rtimes J|/|F| : D \models \Sigma,\ |F| \geq 1\} = \infty$; the factor by which a tuple is multiplied at a hop equals the number of rows of the child that carry its key value.

Proof. (b)⇒(a) is Theorem 3(i). For (a)⇒(b) and the moreover-part, let $h^* = (P.X \to C^*.Z)$ be a hop that is not covered, and let $M \geq 2$. Let V denote the validity and date attributes; we assume, as in the definitions, that V is disjoint from all key and reference lists. Construct D:

- every table $T \neq C^*$ of J, including F, has a single row in which every attribute has the value 1;
- C^* has M rows $t_1, \ldots, t_M$: t_1 has value 1 in every attribute; for $i \geq 2$, t_i has value 1 in the attributes of $Z \cup V$ and NULL in all other attributes.
- Tables outside J are empty.

Every date and validity bound equals 1, with one exception: if h^* is non-temporal, row t_i of C^* gets the validity interval $[i, i]$ (validity attributes of C^* only; dates stay 1). Hence the t_i have pairwise disjoint intervals, every temporal predicate involving t_1 or a table other than C^* holds, and a temporal key atom $\mathsf{U}(C^*, W; fr, to)$ holds for every W. We check $D \models \Sigma$. *Key atoms on $T \neq C^*$* hold because T has at most one row. *Key atoms $\mathsf{U}(C^*, W)$ or $\mathsf{U}(C^*, W; fr, to)$ in Σ*: since h^* is not covered, either the atom is temporal while h^* is not, and then it holds by the choice of intervals above, or $W \not\subseteq Z$. In the latter case W contains an attribute outside $Z \cup V$, in which $t_2, \ldots, t_M$ are NULL. Those rows are exempt, and only t_1 remains. *Inclusion atoms $\mathsf{I}(R.X' \subseteq S.Y')$ in Σ*: every row of every non-empty table other than C^* is constant 1, and so is t_1. If $R \neq C^*$, the single row of R has X'-value $(1, \ldots, 1)$, which is matched by the constant-1 row of S (by t_1 if $S = C^*$). If $R = C^*$, then t_1 is matched likewise; for $i \geq 2$, either $X' \subseteq Z \cup V$, and then $t_i[X'] = (1, \ldots, 1)$ is matched as well, or $t_i[X']$ is not total and t_i is exempt. Temporal inclusion atoms are handled identically, because all dates are 1 and t_1, like every row outside C^*, has the interval $[1, 1]$.

Now evaluate $F \rtimes J$. Every row reaching P along the path from F is the constant-1 row of P (all hops above h^* match exactly one constant row). At h^* it matches $t_1, \ldots, t_M$, because each $t_i[Z] = (1, \ldots, 1) = p[X]$ and every temporal condition holds. Below C^* every tuple produces exactly one tuple by Lemma 2 (every other table has at most one row). Hence $|F \rtimes J| = M > 1 = |F|$, which proves (a)⇒(b), and letting $M \to \infty$ gives the moreover-part. The multiplication factor at a hop is $\max(1, |M_i(q)|)$ by Lemma 2, which is the number of child rows that carry the key value $q[P][X_h]$ (and satisfy the temporal condition). □

Corollary 6 (Reverse hops)**.** *If Σ consists of the atoms of the candidate edges used by J, then J is safe on all instances satisfying Σ if and only if no hop traverses its edge in reverse, unless the child of that reverse hop is also entered, by another edge of J, through a list $W \subseteq Z_h$. In particular, a* 1:1 *relationship that is to be traversed in both directions needs two key certificates, $\mathsf{U}(S, Y)$ and $\mathsf{U}(R, X)$.*

Proof. A forward hop over e is covered by $\mathsf{U}(S_e, Y_e) \in \Sigma$. A reverse hop over e enters $R_e.X_e$, and Σ contains a key atom on R_e only if another edge of J enters R_e, which is the stated exception. Apply Theorem 5. □

Proposition 7 (The fan-out statistic is one-directional)**.** *For a candidate edge $e = (R.X \to S.Y)$ let $\mathrm{fo}(e) = |R \rtimes_{\theta_e} S|/|R|$. Then $\mathrm{fo}(e) = 1$ if and only if every value x that occurs as a total $r[X]$ in R is carried by at most one row of S, i.e. $\mathsf{U}(S, Y)$ holds on the rows of S whose key is referenced by R. The statistic places no restriction on the reverse traversal $S \to R$: for every M there is an instance with $\mathrm{fo}(e) = 1$, $\mathsf{U}(S, Y)$ and $\mathsf{I}(R.X \subseteq S.Y)$ true, and $|S \rtimes_{\theta_e} R| = M|S|$.*

Proof. By Lemma 2, $|R \rtimes S| = \sum_{r \in R} \max(1, m(r))$, where $m(r)$ is the number of rows $s \in S$ with $\theta_e(r, s)$. The sum equals $|R|$ if and only if $m(r) \leq 1$ for all r, which is the stated condition. For the second claim, let S have one row with $Y = 1$ and R have M rows with $X = 1$: then $\mathrm{fo}(e) = 1$ and both atoms of e hold, while the reverse join produces M rows from the single row of S. □

4 Hardness of budgeted certification (paper: Theorem `thm:hard`)

Recall Densest-k-Subgraph (DkS): given a graph $G = (V, E)$ and an integer k, find $K \subseteq V$ with $|K| = k$ maximizing $|E(G[K])|$, the number of edges with both endpoints in K [7]. Its decision version contains k-Clique (ask for $\binom{k}{2}$ edges), which is NP-complete [9].

Theorem 8 (Hardness)**.** *Offline BJC is NP-hard, even when the schema is a star (one fact table, every candidate edge goes from the fact table to a dimension table), every key atom has cost* 0 *(declared keys), every inclusion atom has cost* 1*, every query has weight* 1 *and requires exactly two dimension tables, every atom is true, and the rules $\mathcal{R}$ derive nothing. The reduction preserves objective values exactly. If, instead, costs are rows read ($c(\mathsf{U}(S, Y)) = |S|$, $c(\mathsf{I}(R.X \subseteq S.Y)) = |R| + |S|$), the statement continues to hold.*

Proof. Given a DkS instance $(G = (V, E), k)$, build one fact table F with a reference attribute x_v for each $v \in V$, and one dimension table D_v with a key attribute id for each $v \in V$. The candidate edges are $e_v = (F.x_v \to D_v.id)$. To realize the instance, let F have one row and D_v one row each, all values equal to 1; then all atoms hold. For each $\{u, v\} \in E$ the workload contains $q_{uv} = (F, \{D_u, D_v\}, 1)$. Atom costs: $c(\mathsf{U}(D_v, id)) = 0$ and $c(\mathsf{I}(F.x_v \subseteq D_v.id)) = 1$. The budget is $B = k$. $\mathcal{R}$ derives no inclusion atom: transitivity needs an inclusion atom whose left side is some $D_v.id$, and none exists. The superkey rule derives only key atoms, which are free anyway. For a set P of probed inclusion atoms, let $K(P) = \{v : \mathsf{I}(F.x_v \subseteq D_v.id) \in P\}$. Because the schema is a star, q_{uv} is certified exactly when e_u and e_v are certified, that is, when $u, v \in K(P)$. The certified weight is therefore $|E(G[K(P)])|$, and $|K(P)| = c(P) \leq k$. Conversely, every $K \subseteq V$ with $|K| \leq k$ is realized by a feasible P. Since adding vertices never removes induced edges, the optimum of BJC equals the DkS optimum, and solutions map to solutions of equal value in both directions in polynomial time. With $K = \binom{k}{2}$ this decides k-Clique. For the row-count cost model, each $\mathsf{U}(D_v, id)$ costs 1 and each I costs 2. A query needs both atoms of both its edges, so a vertex costs 3 and the budget $B = 3k$ yields the same correspondence. □

Remark 9. All priors may be set to 1, so the hardness holds for the adaptive (online) problem as well: with all atoms certainly true, adaptivity gives no advantage. The decision version of offline BJC is in NP when $\mathrm{Cl}_{\mathcal{R}}$ is polynomial-time computable (true for the rules of $\mathcal{R}$), so it is NP-complete.

Corollary 10 (Inapproximability)**.** *On the instances of Theorem 8 (with $n = |V|$ dimension tables):*

(i) offline BJC has no PTAS unless $\mathrm{NP} \subseteq \bigcap_{\epsilon>0} \mathrm{BPTIME}(2^{n^\epsilon})$ *[10];*

(ii) assuming the Exponential Time Hypothesis, there is a constant $c > 0$ such that no polynomial-time algorithm approximates offline BJC within a factor $n^{1/(\log\log n)^c}$ [11].

Proof. The reduction of Theorem 8 maps every solution of the BJC instance to a DkS solution of the same value and has the same optimum, so an α-approximation for BJC yields an α-approximation for DkS on n-vertex graphs. Khot [10] shows that DkS has no PTAS under the stated assumption, and Manurangsi [11] shows the stated ETH-hardness of approximating DkS (with perfect completeness). □

Remark 11. The best known polynomial-time approximation for DkS has ratio $O(n^{1/4+\epsilon})$ [1]. We make no positive approximation claim for BJC.

Corollary 12 (Neither submodular nor supermodular)**.** *Let $f(P)$ be the certified weight of a set P of probed atoms. In general f is neither submodular nor supermodular. In particular, the marginal gain of a single atom can be* 0 *although a pair of atoms has positive gain, so greedy selection of single atoms can stall at value* 0*.*

Proof. Complementarity. One query needs two inclusion atoms a, b (Theorem 8 with one edge uv). Then $f(\emptyset) = f(\{a\}) = f(\{b\}) = 0$ and $f(\{a, b\}) = 1$, so $f(\{a, b\}) - f(\{a\}) = 1 > 0 = f(\{b\}) - f(\emptyset)$, which violates submodularity. *Substitutability.* One query with one target reachable along two alternative single-hop edges whose inclusion atoms are a and b (keys free): $f(\{a\}) = f(\{b\}) = f(\{a, b\}) = 1$ and $f(\emptyset) = 0$, so $f(\{a, b\}) - f(\{a\}) = 0 < 1 = f(\{b\}) - f(\emptyset)$, which violates supermodularity. For the stall: with budget 2, a single-atom greedy that picks by marginal gain sees only zero gains in the first instance. □

5 Tractable case: tree-structured candidate graphs (paper: Theorem `thm:tree`)

Hypotheses.

(H1) every query has a single target, $|D_q| = 1$;

(H2) for every fact table F, the candidate edges reachable from F form an out-tree T_F rooted at F;

(H3) the trees of distinct facts share no atom (in particular no key atom);

(H4) $\mathcal{R}$ derives no atom of any T_F from other candidate atoms (e.g. $\mathcal{R} = \emptyset$ or no relevant superkey or transitivity premises exist); and

(H5) costs are non-negative integers (for the pseudo-polynomial bound only).

Inside a single tree every table has exactly one entering edge, so a tree never shares a key atom with itself.

Tree knapsack. Given a rooted forest with node costs $c_v \geq 0$ and profits $\pi_v \geq 0$, and a budget B, the tree knapsack problem asks for an ancestor-closed set S of nodes (if $v \in S$ then its parent is in S or is a root that is always present) maximizing $\pi(S)$ subject to $c(S) \leq B$.

Theorem 13 (Tree case)**.** *Under (H1)–(H4), offline BJC is equivalent to tree knapsack on a forest with $n \leq |E_c|$ nodes. Consequently it is solvable in $O(nB)$ time under (H5), and it admits an FPTAS with running time $O(n^3/\epsilon)$. Pseudo-polynomial algorithms and an FPTAS for tree-structured precedence-constrained knapsack are due to Johnson and Niemi [8]; we give self-contained versions below.*

Proof. Reduction. For each fact F, delete from T_F every node whose entering edge has a false atom, together with its subtree; those nodes can never be certified (truth is known offline). For each remaining non-root node v of T_F with entering edge e_v, set $c_v = c(\mathsf{U}(S_{e_v}, Y_{e_v})) + c(\mathsf{l}_{e_v})$ and $\pi_v = \sum\{w_q : F_q = F, D_q = \{v\}\}$. Queries with $D_q = \{F_q\}$ are certified at no cost and add a constant. By (H2), a query $(F, \{v\})$ is certified iff all edges on the unique root path to v are certified. By (H3), the atoms of these edges belong to T_F only, and by (H4) they are certified only if probed. Given a feasible probe set P, let $S(P)$ be the set of nodes v all of whose root-path atoms are in P. Then $S(P)$ is ancestor-closed, $c(S(P)) \leq c(P)$ (by (H3) and the uniqueness of the entering edge, each node's atoms are charged once), and $S(P)$ certifies exactly the queries certified by P. Conversely, an ancestor-closed S is realized by probing the atoms of its nodes at cost $c(S)$. Hence the optima coincide, and solutions translate in both directions.

Dynamic program. Add one virtual root above all fact roots, and let $v_1, \ldots, v_n$ be the non-fact nodes in preorder of the resulting tree, with $\mathrm{nx}(i)$ the index of the first node after the subtree of v_i ($n+1$ if none). Facts are always present. Define $f(n+1, b) = 0$ and, for $i \leq n$,

$$f(i,b) = \max\big\{\, f(\mathrm{nx}(i), b),\ \ \pi_{v_i} + f(i+1, b - c_{v_i}) \ \text{ if } c_{v_i} \leq b \,\big\}.$$

Claim: $f(i,b)$ is the maximum of $\pi(S)$ over sets $S \subseteq \{v_i, \ldots, v_n\}$ with $c(S) \leq b$ that are *admissible at i*, meaning that every ancestor of a node of S with index $\geq i$ is in S. We use a property of preorder: every ancestor of v_j ($j \geq i$) with index $< i$ is an ancestor of v_i, because preorder subtree intervals are nested or disjoint. Hence admissibility at 1 is exactly ancestor-closedness, and $f(1, B)$ is the optimum. To prove the claim, take an admissible S at i. If $v_i \notin S$, then no descendant of v_i is in S (its ancestor v_i has index $\geq i$), so $S \subseteq \{v_{\mathrm{nx}(i)}, \ldots\}$. Moreover S is admissible at $\mathrm{nx}(i)$, since an ancestor of v_j with index in $[i, \mathrm{nx}(i))$ would make v_j a descendant of v_i. If $v_i \in S$, then $S \setminus \{v_i\}$ is admissible at $i+1$. Conversely, any set admissible at $\mathrm{nx}(i)$ is admissible at i (no node after the subtree has an ancestor inside it), and adding v_i to a set admissible at $i+1$ gives a set admissible at i. Induction on i from $n+1$ down proves the claim. The table has $(n+1)(B+1)$ entries, each computed in $O(1)$ time, which gives $O(nB)$ time.

FPTAS. Remove every node whose root path costs more than B; no feasible solution contains it. Let $\Pi = \max_v \pi_v$ over the remaining nodes. If $\Pi = 0$ the optimum is 0. Otherwise $\mathrm{OPT} \geq \Pi$, because the root path of a node of maximum profit is feasible. Let $\kappa = \epsilon\Pi/n$ and $\pi'_v = \lfloor \pi_v/\kappa \rfloor \leq n/\epsilon$. Run the dual program $g(i,p) =$ minimum cost of a set admissible at i with scaled profit at least p, for $0 \leq p \leq n\lfloor n/\epsilon \rfloor$, using the same recurrence with the roles of cost and profit exchanged. This takes $O(n \cdot n^2/\epsilon)$ time and returns an admissible S' with $c(S') \leq B$ maximizing π'. For an optimal S^*, $\pi(S') \geq \kappa\pi'(S') \geq \kappa\pi'(S^*) \geq \pi(S^*) - n\kappa = \mathrm{OPT} - \epsilon\Pi \geq (1-\epsilon)\mathrm{OPT}$. Note that the scaling is applied to profits: scaling costs would violate the hard budget. □

Remark 14 (Necessity of (H1))**.** The instances of Theorem 8 have a single fact whose candidate graph is a star, which is a tree, so dropping (H1) restores NP-hardness.

Proposition 15 (Shared key atoms break tractability)**.** *Offline BJC remains NP-hard under (H1), (H2) and (H4) when (H3) is dropped, even if every fact's candidate tree has depth* 2*, every key atom costs* 1*, and every inclusion atom costs* 0 *(enforced foreign keys).*

Proof. Reduce from k-Clique. Given $G = (V, E)$ and k, create for every $v \in V$ the tables $A_v(id, c_w : w \in N[v])$ and $B_v(id)$, where $N[v]$ is the closed neighbourhood of v, with candidate edges $A_v.c_w \to B_w.id$. For every ordered pair (u, v) with $\{u, v\} \in E$, and for every pair (v, v), create a fact $F_{uv}(a)$ with the edge $F_{uv}.a \to A_u.id$, and the query $q_{uv} = (F_{uv}, \{B_v\}, w_{uv})$, where $w_{uv} = 1$ for $u \neq v$ and $w_{vv} = M := 2|E| + 1$. Let every table have one row with all values 1, so that all atoms hold. Set the budget $B = 2k$. The tables reachable from F_{uv} are A_u and B_w for $w \in N[u]$, which form a tree of depth 2, so (H1), (H2) and (H4) hold (no inclusion atom has a B-table on its left side, and the superkey rule only yields key atoms that are not needed). The key atoms $\mathsf{U}(A_u, id)$ and $\mathsf{U}(B_v, id)$ are shared among facts. With inclusion atoms free, a probe set corresponds to sets $X = \{u : \mathsf{U}(A_u, id) \text{ probed}\}$ and $Y = \{v : \mathsf{U}(B_v, id) \text{ probed}\}$ with $|X|+|Y| \leq 2k$, and q_{uv} is certified iff $u \in X$ and $v \in Y$. The certified weight is $M|X \cap Y| + |\{(u, v) : \{u, v\} \in E, u \in X, v \in Y\}|$. If K is a k-clique, $X = Y = K$ gives $Mk + k(k-1)$. Conversely, suppose the weight is at least $Mk + k(k-1)$. The second term is at most $2|E| < M$, so $M|X \cap Y| > M(k-1)$ and $|X \cap Y| \geq k$. Together with $|X| + |Y| \leq 2k$ this forces $X = Y =: K$ with $|K| = k$. The second term then equals $2|E(G[K])| \geq k(k-1)$, so K is a clique. □

6 Probe order within a bundle (cited; used as the inner loop)

Proposition 16 (Series-system testing [2, 13])**.** *Let $a_1, \ldots, a_m$ be independent atoms, true with probability p_i, and let $c_i > 0$. Probe them sequentially until one is false or all are true. The expected cost of the order π is $\sum_{i=1}^{m} c_{\pi(i)} \prod_{j<i} p_{\pi(j)}$, and it is minimized by sorting by $c_i/(1-p_i)$ in increasing order (ratio $+\infty$ when $p_i = 1$).*

Proof. This is a classical result. For completeness, here is the adjacent exchange. Swapping a and b at consecutive positions with common prefix probability ρ changes the expected cost from $\rho(c_a + p_a c_b)$ to $\rho(c_b + p_b c_a)$. The former is no larger iff $c_a(1-p_b) \leq c_b(1-p_a)$, i.e. iff $c_a/(1-p_a) \leq c_b/(1-p_b)$. Any order can be sorted by such exchanges without increasing the cost. □

The proposition requires independence. Atoms on the same table may be correlated, in which case the rule is a heuristic.

7 Sampled inclusion certificates (paper: Theorem `thm:sound`(b))

Throughout this section the data are fixed while the samples are drawn, every hop of the tree is forward, and every key atom on the tree holds exactly. Then the maps φ_T of Theorem 3(i) exist.

Definition 17 (Fact-weighted orphan rate)**.** For a hop $h = (P.X \to C.Z)$ of a tree J rooted at F, a row $r \in F$ is *orphaned at h* if $\varphi_P(r) \neq \perp$, $\varphi_P(r)[X]$ is total (and, for temporal hops, $\varphi_P(r).d \neq \mathsf{NULL}$), and $\varphi_C(r) = \perp$. The *fact-weighted orphan rate* is $\rho_h = |\{r \in F : r \text{ orphaned at } h\}|/|F|$. A fact row is *mis-assigned* if it is orphaned at some hop. If $P = F$, then ρ_h is the ordinary orphan rate of $\mathsf{I}(F.X \subseteq C.Z)$.

Sampled test. $\mathrm{Test}_h(m)$ draws m rows of F uniformly at random (independently with replacement, or without replacement), computes φ_P and φ_C for each by key lookups along the path, and accepts iff no sampled row is orphaned at h. For a hop leaving the root this is exactly "sample m rows of $R = F$ and look up their partners".

Theorem 18 (Sampled certificates)**.** *Let $\epsilon, \delta \in (0, 1)$.*

(i) For every hop h, $\Pr[\mathrm{Test}_h(m)$ accepts and $\rho_h > \epsilon] \le (1-\epsilon)^m \le e^{-\epsilon m}$, which is at most δ for $m \ge \ln(1/\delta)/\epsilon$.

(ii) The fraction of mis-assigned fact rows is at most $\sum_h \rho_h$, where hops with an exactly verified inclusion atom contribute 0. Every fact row still occurs exactly once in $F \ltimes J$ (Theorem 3(i)); mis-assigned rows fall into NULL groups.

(iii) (Fixed tree.) If J has k sampled hops, fixed before sampling, and each is tested with $m \ge \ln(k/\delta)/\epsilon$ fresh samples, then with probability at least $1-\delta$ every accepted hop has $\rho_h \le \epsilon$. Hence, if all are accepted, at most a $k\epsilon$ fraction of fact rows is mis-assigned.

(iv) (Adaptive use.) Suppose an adaptive algorithm runs at most N sampled tests in total, where N is fixed in advance (e.g. $N = B$, since every test reads at least one row). If each test uses $m \ge \ln(N/\delta)/\epsilon$ fresh samples, then with probability at least $1-\delta$ every accepted test, whichever tree it ends up in, has $\rho \le \epsilon$. Every certified tree with k sampled hops therefore mis-assigns at most a $k\epsilon$ fraction of its fact rows.

Proof. (i) If $\rho_h \le \epsilon$ the event is empty. Otherwise each draw with replacement avoids the orphaned rows with probability $1 - \rho_h < 1 - \epsilon$, independently, so acceptance has probability $(1-\rho_h)^m < (1-\epsilon)^m$. Without replacement, with N_g non-orphaned rows among $|F|$, acceptance has probability $\prod_{i<m} \frac{N_g - i}{|F| - i} \le (N_g/|F|)^m = (1-\rho_h)^m$. Finally $(1-\epsilon)^m \le e^{-\epsilon m} \le \delta$ when $m \ge \ln(1/\delta)/\epsilon$. (ii) A mis-assigned row is orphaned at some hop, so a union bound over hops gives the bound. A hop whose inclusion atom holds has $\rho_h = 0$ by Theorem 3(iii) applied to that hop. (iii) Apply (i) with δ/k to each hop and take a union bound over the k hops; on the complementary event, (ii) gives at most $k\epsilon$. (iv) Let $\mathcal{E}_t$ be the event that the t-th test accepts although its hop has $\rho > \epsilon$. Conditioned on the history before test t (which determines the hop tested), the fresh samples give $\Pr[\mathcal{E}_t \mid \text{history}] \le \delta/N$ by (i). Hence $\Pr[\bigcup_{t \le N} \mathcal{E}_t] \le \delta$. On the complement, (ii) applies to every certified tree. □

Proposition 19 (Row-level sampling is unsound below the root)**.** *For a hop $h = (P.X \to C.Z)$ with $P \ne F$, an (ϵ, δ) guarantee on the orphan rate of $\mathsf{I}(P.X \subseteq C.Z)$, measured over the rows of P, does not bound the fraction of mis-assigned fact rows. For every $\epsilon > 0$ and every sample size m there are instances in which the orphan rate of P is below ϵ, a row-level test with m uniform draws from P accepts with probability at least $1 - m/|P|$ (arbitrarily close to 1), and yet every fact row is mis-assigned.*

Proof. Let $F \to P \to C$ be a path. P has $n_P > 1/\epsilon$ rows with distinct keys, exactly one of which, p^*, has an X-value without partner in C. Every row of F references p^*. The orphan rate of P is $1/n_P < \epsilon$. A row-level test with m uniform draws from P accepts with probability $(1 - 1/n_P)^m \ge 1 - m/n_P$. Yet $\varphi_C(r) = \perp$ for every $r \in F$, so $\rho_h = 1$. □

Remark 20. Consequently, two guarantees must be distinguished. The *per-atom* guarantee (an accepted $\mathsf{I}(R.X \subseteq S.Y)$ has orphan rate at most ϵ over the rows of R, with probability at least $1 - \delta$) follows from Theorem 18(i) applied to the one-hop tree rooted at R, and it holds for every

sampled atom. The *tree-level* bound on mis-assigned fact rows (Theorem 18(ii)–(iv)) requires either that every sampled hop leaves the anchor table, where the two rates coincide, or that fact rows are sampled and followed along the path (Definition 17). Our implementation samples rows of R for every inclusion atom with $|R| > 10m$, with $m = \lceil \ln(1/\delta)/\epsilon \rceil$ per atom; it therefore provides the per-atom guarantee, and the tree-level bound only for trees whose sampled hops leave the anchor and with δ divided by the number of sampled hops. The uniformity assumption is essential: an engine sampling primitive that is not uniform over rows (e.g. one biased against recently appended row groups) voids (i).

8 Key atoms cannot be certified by sampling (paper: Theorem `thm:lower`)

Model. An algorithm is given $n = |S|$ and accesses S only by reading rows by position, at cost 1 per read (adaptively, any order). No index or other auxiliary information on S is available. It outputs `unique` (certifies $\mathsf{U}(S, Y)$) or `dup`. A randomized algorithm has *error at most* ϵ if on every input its output is wrong with probability at most ϵ.

Theorem 21 (Linear lower bound for key atoms)**.** *Let $n \geq 2$.*

(i) (One-sided.) An algorithm that never outputs `unique` *on an instance violating $\mathsf{U}(S, Y)$ reads all n rows in every run in which it outputs* `unique`*. If it outputs* `unique` *on a key-satisfying instance with probability β, its expected number of reads on that instance is at least βn.*

(ii) (Two-sided, worst case.) Every randomized algorithm with error at most $\epsilon < 1/2$ that always reads at most t rows satisfies $t \geq (1 - 2\epsilon)(n - 1)$.

(iii) (Two-sided, expectation.) Every randomized algorithm with error at most $\epsilon < 1/2$ reads at least $(1 - 2\epsilon)^2(n - 1)/8$ rows in expectation on some input.

The hard instances differ from a key-satisfying instance in a single row, i.e. they contain a single duplicate pair.

Proof. (i) Fix a run (a random string) that outputs `unique` on a key-satisfying instance x and leaves position j unread. Let x' be x with row j replaced by a copy of row $i \neq j$ in the attributes Y (with Y total). On x' the run reads the same contents, by induction over its reads, since only position j differs. So it outputs `unique` on x', which violates the key. The expectation bound follows. (ii) We use Yao's minimax principle [14] in its easy direction. Let x^0 have Y-values $1, 2, \ldots, n$, and for $2 \leq j \leq n$ let x^j equal x^0 except that row j has Y-value 1. Let μ put mass $1/2$ on x^0 and $1/(2(n-1))$ on each x^j. Let A be a deterministic algorithm reading at most t rows, and let Q be the set of positions it reads on x^0, so $|Q| \leq t$. For $j \notin Q$, A behaves identically on x^j and x^0. If $A(x^0) =$ `dup`, A errs with μ-probability at least $1/2$. Otherwise it errs on every x^j with $j \in \{2, \ldots, n\} \setminus Q$, a set of at least $n - 1 - t$ positions, so the error is at least $(n - 1 - t)/(2(n-1))$. In both cases $\mathrm{err}_\mu(A) \geq \frac{1}{2}(1 - \frac{t}{n-1})$. A randomized algorithm $\mathcal{A}$ is a distribution over such A, so $\max_x \Pr[\mathcal{A} \text{ errs on } x] \geq \mathbb{E}_{x \sim \mu} \Pr[\mathcal{A} \text{ errs on } x] = \mathbb{E}_A[\mathrm{err}_\mu(A)] \geq \frac{1}{2}(1 - \frac{t}{n-1})$. Error at most ϵ thus forces $t \geq (1 - 2\epsilon)(n - 1)$. (iii) Let $\mathcal{A}$ have error at most ϵ and expected reads at most T on every input. Let $\gamma = (1 - 2\epsilon)/4$, and let $\mathcal{A}'$ simulate $\mathcal{A}$ but stop with output `unique` before read $\lceil T/\gamma \rceil + 1$. By Markov's inequality the outputs differ with probability at most γ, so $\mathcal{A}'$ has error at most $\epsilon + \gamma < 1/2$ and reads at most $T/\gamma + 1$ rows. By (ii), $T/\gamma + 1 \geq (1 - 2\epsilon - 2\gamma)(n-1) = \frac{1-2\epsilon}{2}(n-1)$, so $T \geq \gamma\big(\frac{1-2\epsilon}{2}(n-1) - 1\big)$, which is $(1 - 2\epsilon)^2(n-1)/8 - O(1)$. □

Remark 22 (Why a tolerance does not help for keys)**.** One might certify "few duplicates" instead of U. This does not bound aggregation error: in x^j a single duplicated key, if referenced by every fact row, doubles every `COUNT` and `SUM`. In contrast, a violation of an inclusion atom at rate ρ is witnessed by single fact rows, and their number is exactly what bounds the mis-assignment (Theorem 18). An exact key probe by hashing or sorting reads n rows, matching the bound. A declared and enforced `UNIQUE` constraint is precisely an auxiliary structure outside this model, which is why declared keys are free.

9 Incremental maintenance (paper: Theorem `thm:maint`)

Update model. A batch ΔD deletes a set Δ_T^- of existing rows from, and inserts a set Δ_T^+ of new rows into, each table T. An update of a row is a deletion followed by an insertion. A new SCD version is an insertion, possibly together with an update that closes the previous version. We write $|\Delta D| = \sum_T |\Delta_T^-| + |\Delta_T^+|$ and $D \oplus \Delta D$ for the result.

Lemma 23 (Monotonicity)**.** *(i) A key atom $\mathsf{U}(S, Y)$ or $\mathsf{U}(S, Y; fr, to)$ true on D can be false on $D \oplus \Delta D$ only if $\Delta_S^+ \neq \emptyset$. Deletions from S and changes to other tables cannot falsify it.*

(ii) An inclusion atom $\mathsf{I}(R.X \subseteq S.Y)$ or $\mathsf{I}(R.X \subseteq S.Y \,@\, d)$ true on D can be false on $D \oplus \Delta D$ only if $\Delta_R^+ \neq \emptyset$ or $\Delta_S^- \neq \emptyset$. Deletions from R, insertions into S and changes to other tables cannot falsify it.

In particular an update that closes a version (shrinks its interval) is a deletion from S and can falsify a temporal inclusion atom. A new version is an insertion into S and can falsify a temporal key atom.

Proof. (i) U is a universal statement over pairs of rows of S. After the batch, every pair of rows of S that does not involve an inserted row was already a pair before, so a violating pair must contain an inserted row. (ii) I has the form $\forall r \in R\, \exists s \in S$. If no row is inserted into R, every row of R after the batch was present before. If no row is deleted from S, every partner it had is still present. Hence the atom still holds. □

Auxiliary state. For every pair (S, Y) that carries a maintained key atom or is the target of a maintained inclusion atom:

- a hash map $\mathrm{cnt}_{S,Y} : y \mapsto |\{s \in S : s[Y] = y\}|$ over total values y (entries with count 0 are removed), and a counter $\mathrm{dup}_{S,Y} = |\{y : \mathrm{cnt}_{S,Y}(y) \geq 2\}|$.

For every maintained inclusion atom $a = \mathsf{I}(R.X \subseteq S.Y)$:

- a hash map $\mathrm{ref}_a : x \mapsto |\{r \in R : r[X] = x\}|$ over total x, and a counter, the number of orphan rows,
$$\mathrm{orph}_a = \textstyle\sum_{x:\, \mathrm{cnt}_{S,Y}(x)=0} \mathrm{ref}_a(x).$$

The *invariant* is that every stored quantity equals its definition on the current database. Then $\mathsf{U}(S, Y)$ holds iff $\mathrm{dup}_{S,Y} = 0$, I holds iff $\mathrm{orph}_a = 0$, and the exact orphan rate is $\mathrm{orph}_a/|R|$.

Per-row update rules. Inserting a row s with total $y = s[Y]$ into S: increment $\mathrm{cnt}(y)$; if it becomes 2, increment dup; if it becomes 1, then for each inclusion atom a into (S, Y) subtract $\mathrm{ref}_a(y)$ from orph_a. Deleting s: decrement $\mathrm{cnt}(y)$; if it drops from 2 to 1, decrement dup; if it drops to 0, add $\mathrm{ref}_a(y)$ to orph_a for each a into (S, Y). Inserting a row r with total $x = r[X]$ into R: for each a with left side (R, X), increment $\mathrm{ref}_a(x)$, and increment orph_a if $\mathrm{cnt}_{S,Y}(x) = 0$. Deleting r: the same with decrements. Rows with non-total keys are ignored. Every rule preserves the invariant, which can be checked case by case against the definitions. For instance, when $\mathrm{cnt}(y)$ goes from 0 to 1, the $\mathrm{ref}_a(y)$ rows referencing y stop being orphans and nothing else changes.

Temporal atoms. For a temporal key atom keep, per total key y, a balanced search tree over the validity intervals of the rows with $s[Y] = y$, augmented to maintain the maximum number of intervals sharing a point (standard interval augmentation [5]), and a counter of keys whose maximum is at least 2. For a temporal inclusion atom keep, per key x, an order-statistic tree over the dates $r.d$ of referencing rows, each annotated with its *coverage* (the number of intervals of S-rows with key x that contain it), supporting range-add of coverage and counting of zero-coverage points (the coverage of a newly inserted date is a stabbing count, obtained from two order-statistic trees over the interval endpoints), together with the global orphan counter. Each insertion or deletion of an interval or date updates these structures in $O(\log N)$ time, where N is the number of rows sharing the key.

Theorem 24 (Exact incremental maintenance)**.** *Let $\mathcal{A}^*$ be the maintained atoms and let α be the maximum, over tables T, of the number of atoms of $\mathcal{A}^*$ in which T occurs. Given the auxiliary state for D, processing a batch ΔD with the rules above yields the exact truth value of every atom of $\mathcal{A}^*$ on $D \oplus \Delta D$, in both directions (an atom may become false or true again). The cost is*

- *$O(\alpha|\Delta D|)$ expected amortized time for non-temporal atoms, using hashing with universal hash functions [3], and $O(\alpha|\Delta D| \log |D|)$ deterministic time with balanced search trees instead;*
- *$O(\alpha|\Delta D| \log N)$ time for temporal atoms;*

and no row of D outside ΔD is read. The state uses $O(\#\text{distinct total keys of } S.Y)$ words per (S, Y) and $O(\#\text{distinct total values of } R.X)$ words per inclusion atom for non-temporal atoms, and $O(|S| + |R|)$ words for temporal atoms. Building the state costs one exact verification. If certificates are updated together with the commit of each batch, no query is certified while one of its atoms is false (zero exposure).

Proof. Correctness follows from the invariant, which holds initially by construction and is preserved by each per-row rule, irrespective of the order in which the rows of the batch are processed, since the stored quantities are functions of the final state only. Each changed row touches at most α atoms, and each touch performs $O(1)$ dictionary operations and counter updates (or $O(\log N)$ tree operations). Expected $O(1)$ time per dictionary operation follows from universal hashing with table doubling [3]. The space bounds count one map entry per distinct key. Zero exposure holds because the truth value used by the planner after each commit is the exact one. □

Remark 25 (Boundedness)**.** The cost depends on $|\Delta D|$ and on the schema constant α only, not on $|D|$. In the terminology of Ramalingam and Reps [12], the algorithm is *bounded*: its cost is polynomial in the size of the change to the input, which is at most the size of CHANGED. Hence it is also *relatively bounded* with respect to the exact batch probe in the sense of Fan, Hu and Tian [6], whose cost measure is polynomial in $|\Delta|$ and in the size of the affected area. As usual, the one-time construction of the auxiliary state is not charged to the incremental cost, and the strict reading of

boundedness applies to the expected-time hashing variant. The deterministic variant adds a $\log |D|$ factor.

Remark 26 (Implementation)**.** Lemma 23 alone already gives an exact "check only risky changes" policy, but without auxiliary state the check after a deletion from S needs a scan of R (Proposition 27). Our implementation (`code/maint.py`) realizes exactly these structures: a sorted base array of key counts per key atom and per referenced column list, plus a sorted delta array that absorbs each batch; a reference-count multiset per inclusion atom; and an orphan counter per inclusion atom that is updated from the keys whose count in S crosses zero and from the inserted and deleted references, so that an inclusion atom becomes true again when its orphans disappear. Lookups are binary searches over the sorted arrays, a logarithmic factor over the hash maps assumed above.

Proposition 27 (Memoryless incremental checking needs linear reads)**.** *Consider an algorithm that knows that* $\mathsf{I}(R.X \subseteq S.Y)$ *and* $\mathsf{U}(S, Y)$ *hold on* D*, receives* ΔD*, may read* S *freely, and may read rows of* R *by position at cost* 1 *each, but has no other state. Let* $n = |R|$*. Deciding* I *on* $D \oplus \Delta D$ *after the deletion of a single row of* S *with error at most* $\epsilon < 1/2$ *requires at least* $(1 - 2\epsilon)n$ *reads of* R *in the worst case (and* $\Omega((1 - 2\epsilon)^2 n)$ *in expectation). Symmetrically, deciding* $\mathsf{U}(S, Y)$ *after the insertion of a single row into* S *without state requires* $(1 - 2\epsilon)|S|$ *reads of* S*.*

Proof. Let S have rows with keys $1, 2$, and let the batch delete the row with key 1. Let x^0 be the instance in which every row of R has $X = 2$, and let x^j $(1 \le j \le n)$ be the instance in which row j has $X = 1$ and all other rows have $X = 2$. All of these satisfy I and U before the batch. After it, I holds on x^0 and fails on every x^j. The algorithm's view of S and of ΔD is identical across them. The argument of Theorem 21(ii)–(iii), with the n positions of R in place of $n - 1$, gives the bounds. For U, let the batch insert a row with key 0, let x^0 have the distinct keys $1, \ldots, |S|$, and let x^j differ from x^0 only in that row j has key 0. All instances satisfy U before the batch, and afterwards U fails exactly on the x^j. □

Proposition 28 (Space lower bound)**.** *Let* $\sigma(D)$ *be any auxiliary state such that, for every single-row deletion from* S*, the truth of* $\mathsf{I}(R.X \subseteq S.Y)$ *afterwards is determined by* $\sigma(D)$*,* S *and the deletion, without reading* R*. Then for every* S *with* s *rows and pairwise distinct total keys,* σ *takes at least* 2^s *distinct values as* R *ranges over instances satisfying* I*. Hence some state has at least* s *bits. The hash-map state uses* $O(s)$ *words, which is optimal up to the word size.*

Proof. For $K \subseteq \mathrm{keys}(S)$ let R_K have one row per key in K. Then I holds on (R_K, S), and after deleting the row with key y it holds iff $y \notin K$. The answers to the s possible deletions determine K, so $K \mapsto \sigma(R_K, S)$ is injective. □

10 Path equivalence (paper: Lemma `lem:patheq`)

Let J, J' be two join trees rooted at F that are both safe on D, i.e. every hop is covered by key atoms that hold on D, and that both contain a set $\mathcal{D}$ of *target* tables. Let φ_d, φ'_d be their maps (Theorem 3). For a target d, the root paths p_d in J and p'_d in J' are *equivalent*, written $\mathsf{C}(p_d \equiv p'_d)$, if $\varphi_d = \varphi'_d$ as functions $F \to d \cup \{\bot\}$, i.e. every fact row reaches the same row of d along both paths, or $\bot$ along both. If $p_d = p'_d$ (the same sequence of hops), then $\varphi_d = \varphi'_d$ trivially.

Planner query class. A query $Q = (\sigma, A, \mathrm{agg}, \mu)$ consists of an arbitrary selection predicate σ on the attributes of F (including `IS NOT DISTINCT FROM` tests), a grouping attribute A of some target $d \in \mathcal{D}$, and an aggregate as in Theorem 3(ii). Its answer $Q(J)$ is the set of (group, value)

pairs of `SELECT` $d.A$, `agg(`μ`)` `FROM` $F \rtimes J$ `WHERE` σ `GROUP BY` $d.A$. (Grouping by several target attributes is also covered by the "if" direction below.)

Lemma 29. *For each safe tree, $\varphi_T(r)$ depends only on the attribute values of r.*

Proof. Induction along the root path: each hop's match is determined by the values of the parent row. □

Theorem 30 (Answer invariance iff path equivalence)**.** *Suppose J and J' differ only on the root paths of the targets in $\mathcal{D}' \subseteq \mathcal{D}$. Then $Q(J) = Q(J')$ for every query Q of the planner class if and only if $\mathsf{C}(p_d \equiv p'_d)$ holds for every $d \in \mathcal{D}'$. The "only if" direction is witnessed by a single `COUNT(*)` query.*

Proof. (If.) For every $d \in \mathcal{D}$, $\varphi_d = \varphi'_d$ (by hypothesis for $\mathcal{D}'$, trivially otherwise). By Theorem 3(ii), the value of every group in $Q(J)$ is $\mathrm{agg}\{\{r[\mu] : \sigma(r),\ \varphi_d(r)[A] = g\}\}$, with $\bot[A] = \mathsf{NULL}$. This depends on J only through φ_d, so $Q(J) = Q(J')$, and the same holds for any list of target grouping attributes. (Only if.) Let $d \in \mathcal{D}'$ and $r_0 \in F$ with $t = \varphi_d(r_0) \neq t' = \varphi'_d(r_0)$. Let σ be the conjunction of $F.B$ `IS NOT DISTINCT FROM` $r_0[B]$ over all attributes B of F. The selected rows agree with r_0 on all attributes, so by Lemma 29 they all have images t under J and t' under J'. We claim that some attribute A of d separates t and t' as `GROUP BY` values, where `NULL` forms its own group. If $t = \bot$, then t' was matched through a total list Z' (the entering list of the last hop of p'_d), so any $A \in Z'$ has $t'[A] \neq \mathsf{NULL} = \bot[A]$, and symmetrically if $t' = \bot$. If both are rows, suppose they agreed on all attributes. Then $t[Z] = t'[Z]$ is total for the entering list Z of the last hop of p_d, and in particular $t[W] = t'[W]$ is total for the covering list $W \subseteq Z$. Since $t \neq t'$, this contradicts the key atom $\mathsf{U}(d, W)$, which holds because J is safe. Now the query $(\sigma, A, \texttt{COUNT(*)})$ returns the single group $t[A]$ under J and the single group $t'[A] \neq t[A]$ under J', so the answers differ. □

Corollary 31. *(i) C is an equivalence relation on the safe root paths from F to d (it is equality of functions). Hence k such paths are all equivalent iff $\mathsf{C}(p_1 \equiv p_i)$ holds for $i = 2, \ldots, k$: $k-1$ probes certify all of them, and a single failed probe refutes "all equivalent".*

(ii) Exact certification of $\mathsf{C}(p \equiv p')$ needs $\Omega(|F|)$ reads of fact rows in the worst case, even if all dimension tables are read for free: in the row-read model of Section 8, error at most $\epsilon < 1/2$ requires at least $(1 - 2\epsilon)|F|$ reads.

(iii) One-sided sampling applies to C as to inclusion atoms: if the divergence rate *$|\{r : \varphi_d(r) \neq \varphi'_d(r)\}|/|F|$ exceeds ϵ, then $m \geq \ln(1/\delta)/\epsilon$ uniform fact rows reveal a divergent row with probability at least $1 - \delta$.*

Proof. (i) is immediate. (ii) Let p be the hop $F.a \to d.id$, and let p' be $F.b \to M.id$ followed by $M.dk \to d.id$, where d has keys $1, \ldots, s$ ($s \geq 2$) and M is the identity map ($id = dk$). Let x^0 have $a = b = 1$ in every fact row, and let x^j change row j to $a = 1$, $b = 2$. All key and inclusion atoms hold in all instances, C holds on x^0 and fails on each x^j, and the instances differ in a single fact row. The argument of Theorem 21(ii) with $|F|$ positions applies. (iii) is the computation of Theorem 18(i) with divergent rows in place of orphaned rows. □

Remark 32. The plan's further claim on the incremental maintenance of C (via multisets of first-hop key pairs) is not proved here and is not claimed in the paper.

References

[1] Aditya Bhaskara, Moses Charikar, Eden Chlamtáč, Uriel Feige, and Aravindan Vijayaraghavan. Detecting high log-densities: An $O(n^{1/4})$ approximation for densest k-subgraph. In *Proceedings of the 42nd ACM Symposium on Theory of Computing (STOC)*, pages 201–210, 2010.

[2] Richard Butterworth. Some reliability fault-testing models. *Operations Research*, 20(2):335–343, 1972.

[3] J. Lawrence Carter and Mark N. Wegman. Universal classes of hash functions. *Journal of Computer and System Sciences*, 18(2):143–154, 1979.

[4] Marco A. Casanova, Ronald Fagin, and Christos H. Papadimitriou. Inclusion dependencies and their interaction with functional dependencies. In *Proceedings of the 1st ACM SIGACT-SIGMOD Symposium on Principles of Database Systems (PODS)*, pages 171–176, 1982.

[5] Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein. *Introduction to Algorithms*. MIT Press, Cambridge, MA, 3rd edition, 2009.

[6] Wenfei Fan, Chunming Hu, and Chao Tian. Incremental graph computations: Doable and undoable. In *Proceedings of the 2017 ACM International Conference on Management of Data (SIGMOD)*, pages 155–169, 2017.

[7] Uriel Feige, David Peleg, and Guy Kortsarz. The dense k-subgraph problem. *Algorithmica*, 29(3):410–421, 2001.

[8] David S. Johnson and K. A. Niemi. On knapsacks, partitions, and a new dynamic programming technique for trees. *Mathematics of Operations Research*, 8(1):1–14, 1983.

[9] Richard M. Karp. Reducibility among combinatorial problems. In Raymond E. Miller and James W. Thatcher, editors, *Complexity of Computer Computations*, pages 85–103. Plenum Press, New York, 1972.

[10] Subhash Khot. Ruling out PTAS for graph min-bisection, dense k-subgraph, and bipartite clique. *SIAM Journal on Computing*, 36(4):1025–1071, 2006.

[11] Pasin Manurangsi. Almost-polynomial ratio ETH-hardness of approximating densest k-subgraph. In *Proceedings of the 49th Annual ACM SIGACT Symposium on Theory of Computing (STOC)*, pages 954–961, 2017.

[12] G. Ramalingam and Thomas Reps. On the computational complexity of dynamic graph problems. *Theoretical Computer Science*, 158(1–2):233–277, 1996.

[13] Tonguç Ünlüyurt. Sequential testing of complex systems: A review. *Discrete Applied Mathematics*, 142(1–3):189–205, 2004.

[14] Andrew Chi-Chih Yao. Probabilistic computations: Toward a unified measure of complexity. In *18th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 222–227, 1977.